

GOSFORD HIGH SCHOOL



2013 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics

General Instructions

- Reading Time - 5 minutes
- Working Time - 3 hours
- Write using a blue or black pen. Black pen is preferred
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-16

Total marks (100)

Section I

Total marks (10)

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II

Total marks (90)

- Attempt questions 11 – 16
- Answer in the answer booklets provided, unless otherwise instructed
- Start a new booklet for each question
- All necessary working should be shown for every question
- Allow about 2 hours 45 minutes for this section

SECTION I

- 10 Marks
- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided

Question 1.

5.09674 correct to 3 significant figures is:

- (A) 5.09 (B) 5.10 (C) 5.096 (D) 5.097

Question 2.

Which of the following is equivalent to $\frac{1}{\sqrt{7} + 2\sqrt{3}}$?

- (A) $7 - 2\sqrt{3}$ (B) $7 + 2\sqrt{3}$
(C) $\frac{\sqrt{7} - 2\sqrt{3}}{-5}$ (D) $\frac{\sqrt{7} + 2\sqrt{3}}{-5}$

Question 3.

$\frac{x^2 - 5xy}{x^2 - 25y^2}$ simplifies to:

- (A) $\frac{x}{x - 5y}$ (B) $\frac{x}{x + 5y}$
(C) $\frac{1 - x}{1 - 5y}$ (D) $\frac{x - 5y}{x + 25y}$

Question 4.

Two regular dice are thrown. What is the probability that the throw will **not** result in a double?

- (A) $\frac{1}{36}$ (B) $\frac{31}{36}$ (C) $\frac{35}{36}$ (D) $\frac{5}{6}$

Question 5.

Convert 84° into radians correct to 2 decimal places.

- (A) 2.93 (B) 1.46
(C) 1.47 (D) 4.81

Question 6.

The graph of the function $y = 3\sin 2x$ has:

- (A) amplitude 3 and period π
- (B) amplitude 2 and period $\frac{2\pi}{3}$
- (C) amplitude 3 and period $\frac{\pi}{2}$
- (D) amplitude $\frac{3}{2}$ and period π

Question 7.

What is the equation of the parabola with vertex $(4,2)$ and focus $(3,2)$?

- (A) $(x-4)^2 = 4(y-2)$ (B) $(x-4)^2 = -4(y-2)$
(C) $(y-2)^2 = 4(x-4)$ (D) $(y-2)^2 = -4(x-4)$

Question 8.

$$\frac{d}{dx}(\sin 2x) =$$

- (A) $2 \sin 2x$ (B) $2 \cos 2x$
(C) $-2 \cos 2x$ (D) $\frac{1}{2} \cos 2x$

Question 9.

It is known that for a particular quadratic equation, $\alpha + \beta = -\frac{5}{3}$ and $\alpha\beta = \frac{7}{3}$. The quadratic equation could be:

(A) $6x^2 + 10x + 14 = 0$

(B) $3x^2 - 5x + 7 = 0$

(C) $3x^2 + 5x - 7 = 0$

(D) $5x^2 - 7x + 3 = 0$

Question 10.

What is the angle of inclination of the line $3x + 2y = 7$ with the positive direction of the x axis?

(A) $33^\circ 41'$

(B) $56^\circ 19'$

(C) $123^\circ 41'$

(D) $146^\circ 19'$

Section II

- **90 Marks**
- **Attempt Questions 11-16**
- **Allow about 2 hours 45 minutes for this section.**
- **Answer each question in a separate answer booklet.**
- **All necessary working should be shown for each question.**

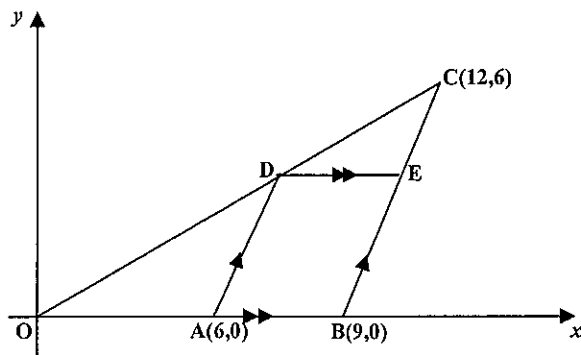
Question 11.

- | | | |
|----|---|---|
| a) | Solve $4^x = 32$ | 2 |
| b) | Fully factorise $40 - 5y^3$ | 2 |
| c) | Solve $ 5 - 2x \geq 9$ | 2 |
| d) | Find the exact value of $\tan \frac{2\pi}{3}$ | 1 |
| e) | Differentiate $\sqrt{x}$ | 1 |
| f) | Find the primitive function of $(x - 4)^6$ | 1 |
| g) | Find $\frac{d}{dx} x \cos(x + 1)$ | 2 |
| h) | State the domain and range of $y = \sqrt{3 - x}$ | 2 |
| i) | Calculate the value of $\log_5 16$, correct to 2 decimal places. | 2 |

Question 12.

Start a new booklet.

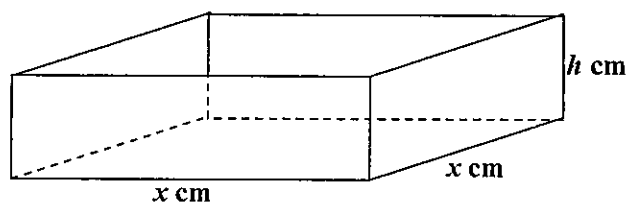
- a) In the diagram, A, B and C are the points (6,0), (9,0) and (12,6) respectively. The equation of the line OC is $x - 2y = 0$. AD is parallel to BC and DE is parallel to the x axis.



- i) Show that the equation of the line AD is $y = 2x - 12$. 2
- ii) Find the coordinates of the point D. 2
- iii) Prove that $\triangle OAD$ is similar to $\triangle DEC$. 2
- b) A total of 300 tickets are sold in a raffle which has 3 prizes. There are 100 red, 100 green and 100 blue tickets. At the drawing of the raffle, winning tickets are NOT replaced before the next draw.
- i) What is the probability that each of the 3 winning tickets is red? 1
- ii) What is the probability that at least one of the winning tickets is not red? 1
- iii) What is the probability that there is one winning ticket of each colour? 2
- c) Find $\int \frac{x}{x^2 + 5} dx$ 1
- d) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$ 2
- e) Evaluate $\int_0^4 \frac{1}{\sqrt{4-x}} dx$ 2

Question 13. Start a new booklet.

- a) A circle with radius 8 cm has an arc PQ 12 cm in length. Find:
- the size of the angle subtended by the arc PQ at the centre of the circle. 1
 - the length of the chord PQ, correct to 2 decimal places. 2
- b) A metal tray, in the shape of a rectangular prism with a square base, is made out of 108 square centimetres of sheet metal. The tray is open at the top. Let x cm be the side length of the base and h cm be the height as shown.



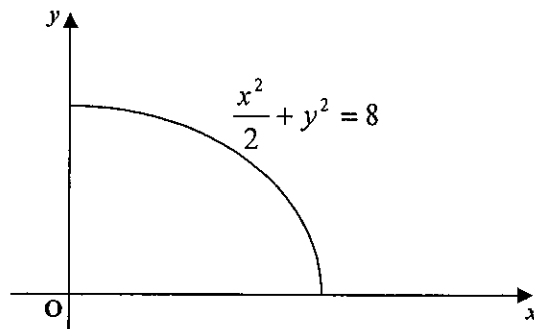
- Show that $h = \frac{108 - x^2}{4x}$ 1
 - Show that the volume, V of the tray is given by $V = 27x - \frac{x^3}{4}$. 2
 - Find the maximum volume of the tray. 3
- c)
- Differentiate $\log_e(\cos x)$ with respect to x . 1
 - Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \tan x dx$ 2
- d)
- Show that $\cos \theta \tan \theta = \sin \theta$. 1
 - Hence solve $8 \sin \theta \cos \theta \tan \theta = \operatorname{cosec} \theta$ for $0 \leq \theta \leq 2\pi$ 2

Question 14. **Start a new booklet.**

- | | | |
|----|--|-------------|
| a) | Solve for x : $e^{2x} + 3e^x - 10 = 0$. | 2 |
| b) | Solve $2 \ln x = \ln(5 + 4x)$. | 2 |
| c) | A(6, 0) and B(1, -3) are fixed points and P(x , y) moves so that PA is perpendicular to PB. Show that the locus of P is
$x^2 + y^2 - 7x + 3y + 6 = 0$ | 3 |
| d) | <p>i) Sketch the graph of $y = 4 \cos 2x$ for $-\pi \leq x \leq \pi$, clearly showing the x and y intercepts.</p> <p>ii) On the same set of axes, sketch the graph of $y = x$.</p> <p>iii) Hence state how many solutions there are to the equation $4 \cos 2x - x = 0$, for $-\pi \leq x \leq \pi$.</p> | 3
1
1 |
| e) | The curve $y = f(x)$ has gradient function $f'(x) = 3x^2 - k$, where k is a constant. | |
| | <p>i) Find the value of k if the curve has a stationary point at (-1, 3).</p> <p>ii) Hence find the equation of the curve.</p> | 1
2 |

Question 15. Start a new booklet.

- a) Consider the function $f(x) = x^4 - 4x^3$
- i) Show that $f'(x) = 4x^2(x - 3)$ 1
 - ii) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 3
 - iii) Sketch the graph of the curve $y = f(x)$, showing the stationary points 2
 - iv) Find the values of x for which the graph of $y = f(x)$ is concave down. 1
- b)



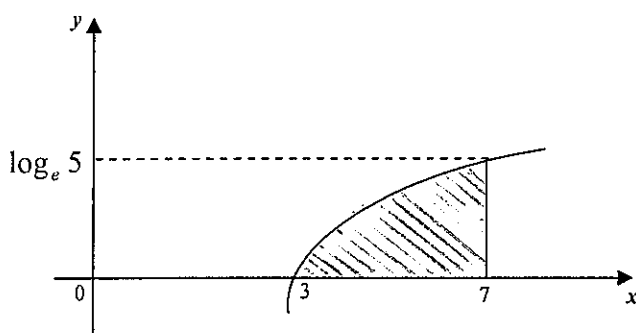
The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is rotated about the x axis. Find the volume of the solid of revolution. 3

- c) Under certain climatic conditions the number N of blue-green algae satisfies the equation $N = Ae^{0.15t}$, where t is measured in days and A is a constant.
- i) Show that the number of algae increases at a rate proportional to the number present. 1
 - ii) When $t = 3$ the number of algae was estimated to be 1.7×10^8 . Evaluate A . 2
 - iii) The number of algae doubles every x days. Find x . 2

Question 16.**Start a new booklet.**

- a) Find the value(s) of k for which the line $kx + y + 1 = 0$ is a tangent to the parabola $y = x^2$. 2
- b) i) Use Simpson's Rule with 3 function values to find an approximation to the area under the curve $y = \frac{1}{x}$ between $x = a$ and $x = 3a$, where a is positive. 2
- ii) Using the result in part i), show that $\ln 3 \approx \frac{10}{9}$. 1

- c) Part of the graph of $y = \log_e(x - 2)$ is shown below.



- i) Find the exact value of the area between the curve and the y axis, bounded by the lines $y = 0$ and $y = \log_e 5$. 3
- ii) Hence find the exact value of the shaded area. 1
- d) A particle is moving in a straight line. Its displacement, x metres, from the origin, O, at time t seconds, where $t \geq 0$, is given by $x = 1 - \frac{7}{t + 4}$.
- i) Find the initial displacement of the particle. 1
- ii) Find the velocity of the particle as it passes through the origin. 2
- iii) Show that the acceleration of the particle is always negative. 1
- iv) Sketch the graph of the displacement of the particle as a function of time. 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

SECTION I

1) (B)

$$2) \frac{1}{\sqrt{7}+2\sqrt{3}} \times \frac{\sqrt{7}-2\sqrt{3}}{\sqrt{7}-2\sqrt{3}}$$

$$= \frac{\sqrt{7}-2\sqrt{3}}{7-12} \quad (C)$$

$$3) \frac{x(\cancel{x-5y})}{(\cancel{x-5y})(x+5y)} \quad (B)$$

4) (D)

$$5) 84 \times \frac{\pi}{180} = 1.47 \quad (C)$$

6) (A)

7) (D)

8) (B)

$$9) x^2 + \frac{5}{3}x + \frac{7}{3} = 0$$

$$3x^2 + 5x + 7 = 0 \quad (A)$$

$$10) 2y = -3x + 7$$

$$y = -\frac{3}{2}x + \frac{7}{2}$$

$$\tan \theta = -\frac{3}{2}$$

$$\theta = 123^\circ 41' \quad (C)$$

SECTION II

$$11) a) 4^x = 32$$

$$2^{2x} = 2^5$$

$$2x = 5$$

$$x = \frac{5}{2}$$

$$b) 40 - 5y^3 = 5(8 - y^3)$$

$$= 5(2 - y)(4 + 2y + y^2)$$

$$c) |5 - 2x| \geq 9$$

$$5 - 2x \geq 9 \quad \text{or} \quad 5 - 2x \leq -9$$

$$2x \leq -4 \quad \quad \quad 2x \geq 14$$

$$x \leq -2 \quad \quad \quad x \geq 7$$

$$d) -\sqrt{3}$$

$$e) \frac{d}{dx} x^{1/2} = \frac{1}{2} x^{-1/2}$$

$$= \frac{1}{2\sqrt{x}}$$

$$f) \frac{(x-4)^7}{7} + C$$

$$g) -x \sin(x+1) + \cos(x+1)$$

$$h) \text{Domain: } x \leq 3$$

$$\text{Range: } y \geq 0$$

$$i) \log_5 16 = \frac{\log 16}{\log 5}$$

$$= 1.72$$

$$12) \text{ a) } M_{BC} = \frac{6}{3}$$

$$i) = 2 \quad \therefore M_{AD} = 2$$

$$y - 0 = 2(x - 6)$$

$$y = 2x - 12$$

$$ii) \quad x - 2y = 0 \quad \text{--- (1)}$$

$$y = 2x - 12 \quad \text{--- (2)}$$

sub (2) into (1)

$$x - 2(2x - 12) = 0$$

$$x - 4x + 24 = 0$$

$$3x = 24$$

$$x = 8$$

$$y = 16 - 12$$

$$= 4$$

$\therefore$ coords of D are (8, 4)

$$iii) \angle CDE = \angle DOA \text{ (corresponding } \angle\text{'s, } DE \parallel OB)$$

$$\angle OAD = \angle ADE \text{ (alternate } \angle\text{'s } DE \parallel OB)$$

$$\text{and } \angle ADE = \angle DEC \text{ (alternate } \angle\text{'s, } AD \parallel BE)$$

$$\therefore \angle OAD = \angle DEC$$

$$\therefore \triangle OAD \parallel \triangle DEC \text{ (equiangular)}$$

$$b) \text{ i) } \frac{100}{300} \times \frac{99}{299} \times \frac{98}{298} = \frac{1617}{44551}$$

$$ii) 1 - \frac{1617}{44551} = \frac{42934}{44551}$$

$$= 0.964$$

$$iii) \frac{100}{300} \times \frac{100}{299} \times \frac{100}{298} \times 6$$

$$= \frac{10000}{44551}$$

$$= 0.224$$

$$c) \int \frac{x}{x^2 + 5} dx = \frac{1}{2} \int \frac{2x}{x^2 + 5} dx$$

$$= \frac{1}{2} \ln(x^2 + 5) + C$$

$$d) \int_0^{\pi/8} \sec^2 2x dx = \left[\frac{1}{2} \tan 2x \right]_0^{\pi/8}$$

$$= \frac{1}{2} \tan \frac{\pi}{4} - 0$$

$$= \frac{1}{2}$$

$$e) \int_0^4 \frac{1}{\sqrt{4-x}} dx$$

$$= \int_0^4 (4-x)^{-1/2} dx$$

$$= \left[-2(4-x)^{1/2} \right]_0^4$$

$$= 4$$

13) a) i) $L = r\theta$

$$12 = 8\theta$$

$$\theta = \frac{3}{2} \text{ radians}$$

ii) $PQ^2 = 8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 1.5$

$$= 128 - 128 \cos 1.5$$

$$= 118.9456382$$

$$\therefore PQ = 10.91 \text{ cm}$$

b) i) $SA = x^2 + 4xh$

$$\therefore x^2 + 4xh = 108$$

$$4xh = 108 - x^2$$

$$h = \frac{108 - x^2}{4x}$$

ii) $V = x^2h$

$$= x^2 \left[\frac{108 - x^2}{4x} \right]$$

$$= \frac{108x - x^3}{4}$$

$$= 27x - \frac{x^3}{4}$$

iii) $\frac{dV}{dx} = 27 - \frac{3x^2}{4}$ $\frac{d^2V}{dx^2} = -\frac{6x}{4}$

for max $\frac{dV}{dx} = 0$ and $\frac{d^2V}{dx^2} < 0$

$$27 - \frac{3x^2}{4} = 0$$

$$3x^2 = 108$$

$$x^2 = 36$$

$$x = \pm 6$$

ignore -ve value as measurement

when $x = 6$, $\frac{d^2V}{dx^2} < 0$

$\therefore$ Max volume when $x = 6$.

$$V = 27(6) - \frac{6^3}{4}$$

$$= 162 - \frac{216}{4}$$

$$= 108 \text{ cm}^3$$

c) i) $\frac{d}{dx} \log(\cos x) = \frac{-\sin x}{\cos x}$

$$= -\tan x$$

ii) $\int_0^{\pi/4} \tan x \, dx = -\left[\ln(\cos x) \right]_0^{\pi/4}$

$$= -\left[\ln\left(\cos \frac{\pi}{4}\right) - \ln(\cos 0) \right]$$

$$= -\ln \frac{1}{\sqrt{2}} + \ln 1$$

$$= -\ln 2^{1/2}$$

$$= -\frac{1}{2} \ln 2$$

d) i) LHS = $\cos \theta \cdot \frac{\sin \theta}{\cos \theta}$

$$= \sin \theta$$

$$= \text{RHS}$$

ii) $8 \sin \theta \cos \theta \tan \theta = \sec \theta$

$$8 \sin^2 \theta = \frac{1}{\sin \theta}$$

$$8 \sin^3 \theta = 1$$

$$\sin^3 \theta = \frac{1}{8}$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

14) a) $e^{2x} + 3e^x - 10 = 0$

let $u = e^x$

$$u^2 + 3u - 10 = 0$$

$$(u+5)(u-2) = 0$$

$$u = -5, 2$$

$$\therefore e^x = -5 \text{ or } e^x = 2$$

no soln

$$x = \ln 2 \\ = 0.693$$

b) $2 \ln x = \ln(5+4x)$

$$\ln x^2 = \ln(5+4x)$$

$$x^2 = 5+4x$$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$x = 5, -1$$

$\therefore x = 5$ only soln.

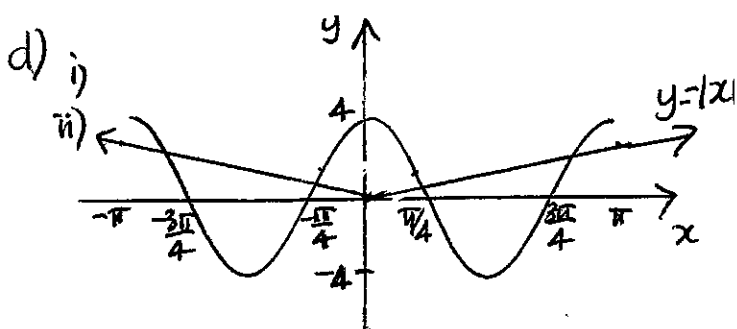
c) $M_{PA} \cdot M_{PB} = -1$

$$M_{PA} = \frac{y}{x-6} \quad M_{PB} = \frac{y+3}{x-1}$$

$$\frac{y}{x-6} \cdot \frac{y+3}{x-1} = -1$$

$$y^2 + 3y = -x^2 + 7x - 6$$

$$x^2 + y^2 - 7x + 3y + 6 = 0$$



iii) 4

e) $f'(x) = 3x^2 - k$

i) $f'(x) = 0$, when $x = -1$

$$3x^2 - k = 0$$

$$3 - k = 0$$

$$k = 3$$

ii) $f'(x) = 3x^2 - 3$

$$f(x) = x^3 - 3x + C$$

when $x = -1, y = 3$

$$3 = -1 + 3 + C$$

$$C = 1$$

$$\therefore f(x) = x^3 - 3x + 1$$

15) a) $f(x) = x^4 - 4x^3$

i) $f'(x) = 4x^3 - 12x^2 \\ = 4x^2(x-3)$

ii) $4x^2(x-3) = 0$

$$4x^2 = 0 \quad x-3 = 0$$

$$x = 0 \quad x = 3$$

when $x = 0, y = 0$

$$x = 3, y = -27$$

$\therefore$ stat. points at $(3, -27)$
and $(0, 0)$

$$f''(x) = 12x^2 - 24x$$

when $x=0$, $f''(x)=0$

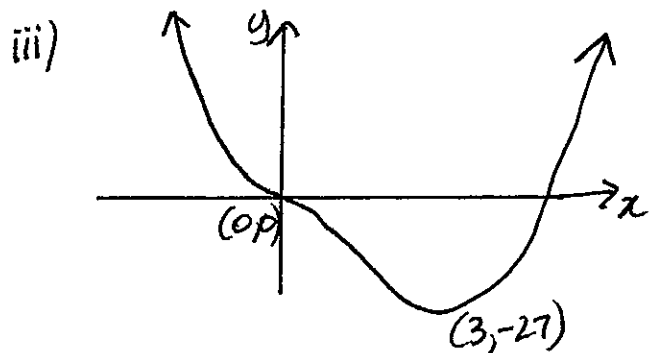
$\therefore$ possible point of inflexion
test for change in concavity

x	-1	0	1
$f''(x)$	+	0	-

$\therefore$ point of inflexion at $(0,0)$

when $x=3$, $f''(x) > 0$

$\therefore$ minimum at $(3, -27)$



iv) $12x^2 - 24x < 0$

olve $12x(x-2) = 0$

$$x = 0, 2$$

$\therefore$ concave down for

$$0 < x < 2$$

b) $\frac{x^2}{2} + y^2 = 8$
 $y^2 = 8 - \frac{x^2}{2}$

$$V = \pi \int y^2 dx$$

$$= \pi \int_0^4 8 - \frac{x^2}{2} dx$$

$$= \pi \left[8x - \frac{x^3}{6} \right]_0^4$$

$$= \pi \left[32 - \frac{64}{6} \right]$$

$$= \frac{64\pi}{3} \text{ cubic units}$$

c) $N = Ae^{0.15t}$

i) $\frac{dN}{dt} = 0.15Ae^{0.15t}$
 $= 0.15N$

$\therefore$ proportional to number present

ii) $1.7 \times 10^8 = Ae^{0.45}$
 $A = \frac{1.7 \times 10^8}{e^{0.45}}$
 $= 1.08 \times 10^8$

iii) $N = (1.08 \times 10^8)e^{0.15t}$

if doubles every x days

then $2 = e^{0.15x}$

$$\ln 2 = 0.15x$$

$$x = 4.62 \text{ days}$$

16) a) $kx + y + 1 = 0$ — (1)
 $y = x^2$ — (2)

from (1) $y = -kx - 1$

$\therefore x^2 = -kx - 1$
 $x^2 + kx + 1 = 0$

eqn has 1 soln if line is a tangent i.e. $\Delta = 0$

$$k^2 - 4 = 0$$

$$k = \pm 2$$

b) i) $A \doteq \frac{a}{3} \left[\frac{1}{a} + \frac{1}{3a} + 4 \cdot \frac{1}{2a} \right]$

$$= \frac{a}{3} \left[\frac{3+1+6}{3a} \right]$$

$$= \frac{10}{9}$$

ii) $\int_a^{3a} \frac{1}{x} dx = \left[\ln x \right]_a^{3a}$

$$= \ln 3a - \ln a$$

$$= \ln 3$$

$$\therefore \ln 3 \doteq \frac{10}{9}$$

c) i) $y = \ln(x-2)$

$$x-2 = e^y$$

$$x = e^y + 2$$

$$A = \int_0^{\ln 5} e^y + 2 dy$$

$$= \left[e^y + 2y \right]_0^{\ln 5}$$

$$= (e^{\ln 5} + 2 \ln 5) - (e^0 + 0)$$

$$= 5 + 2 \ln 5 - 1$$

$$= 4 + 2 \ln 5 \text{ sq units}$$

ii) $A = 7 \ln 5 - (4 + 2 \ln 5)$

$$= 5 \ln 5 - 4 \text{ sq units}$$

d) i) $x = 1 - \frac{7}{t+4}$

When $t=0$, $x = 1 - \frac{7}{4}$

$$= -\frac{3}{4}$$

i.e. $\frac{3}{4}$ m to left of origin

ii) $x = 1 - 7(t+4)^{-1}$

$$\dot{x} = \frac{7}{(t+4)^2}$$

particle passes origin when:

$$0 = 1 - \frac{7}{t+4}$$

$$\frac{7}{t+4} = 1$$

$$7 = t+4$$

$$t = 3 \rightarrow \dot{x} = \frac{7}{(3+4)^2}$$

$$= \frac{1}{7} \text{ m/s}$$

iii) $\ddot{x} = \frac{-14}{(t+4)^3}$

since $t \geq 0$, $(t+4)^3 > 0$

$\therefore \ddot{x}$ always -ve

